

TECHNISCHE UNIVERSITÄT
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AMPL: A Mathematical Programming Language

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Outlines

- **Introduction**
- What is AMPL?
- AMPL features.
- The Syntax
- Run AMPL

- **Examples**
- Example 1: general picture of using AMPL
- Example 2 : AMPL for modeling MPC
- Example 3: advanced example of MPC

- **Installation**

Introduction - 1

- **What is AMPL**

- AMPL is high-level algebraic modelling language to describe, implement and solve the optimization problems.
- AMPL invokes the solvers (optimization algorithms) to find the solution of OP, i.e., AMPL is an interface environment not solving algorithms.

Introduction - 2

- **AMPL features**
  - AMPL syntax is similar to the mathematical notation of optimization problems
  - It supports a lot of solvers:

    CONDOR, CONOPT, Couenne, **CPLEX**, DONLP2, FilMINT, FILTER, MINLP, FortMP, FSQP, Gecode, **Gurobi**, **IPOPT**, JaCoP, KNITRO, LANCELOT, L-BFGS-B, LGO, LINDO, Global, LOQO, LP_SOLVE, MINLP, **MINOS**, MINTO, MOSEK, NPSOL, NSIPS, OOQP, PATH, PCx, PENNON, RAPOSa, SCIP, SNOPT, SOPT, Sulum, TRON, WSAT(OIP), XA, XLSOL, LS-XLSOL, Xpress, ACRS, ALGENCAN, BARON, BLMVM, Bonmin, BPMPD, CBC.

    https://ampl.com/products/solvers/all-solvers-for-ampl/

  - It has IDE (Integrated Development Environment) supports Microsoft Windows, Linux and, macOS.
  - It has API (Application Programming Interface) for Python, R, C++, C#, Java, MATLAB
  - Treats large scale optimization

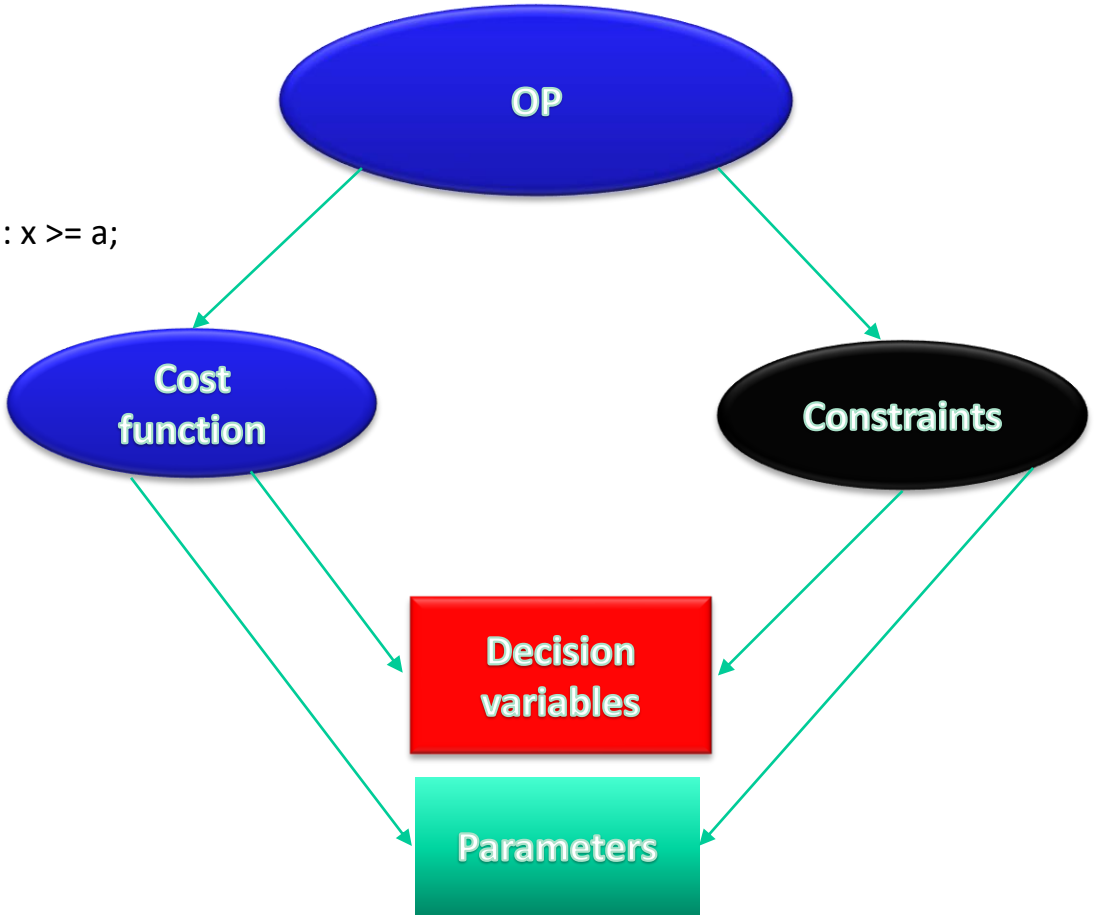
Introduction - 3

$a = 2$

$\underset{x}{minimiz}\ x^2$

$subject\ to\ x \geq a$

```
model;
param a;
var x;
minimize cost: x**2;
s.t.          restriction: x >= a;
data;
 param a := 2;
```



- **Syntax**

- Parameters:

  param parameter_name;

- Variable:

  var variable_name;

- Cost function:

  minimized/maximize cost_function_name:

- Constraints:

  (subject to/s.t.) constrait_name:

Introduction - 4

$a = 2$

$\underset{x}{minimiz}\ x^2$

$subject\ to\ x \geq a$



```
model;
param a;
var x; # the decision variable
minimize cost: x**2;
s.t.          restriction: x >= a;
data;
 param a := 2;
```

- **Syntax**

- Parameters:

  param parameter_name;

- Variable:

  var variable_name;

- Cost function:

  minimize/maximize cost_function_name:

- Constraints:

  (subject to/s.t.) constraint_name:

- **Remarks**

- Every line instruction must be terminated with ";".
- Line comments are preceded by the symbol "#".
- AMPL is "case-sensitive".
- Variable names must be unique

- **AMPL Modes**

- Model mode: this is where we define the parameters, variables, sets, cost function, and constraints.
- Data mode: this is where the data is given a value

Installation – (Install the Program)

1. - Form the webpage: https://ampl.com/try-ampl/download-a-free-demo/

choose the download link based on the operating system (OS) to install the program.

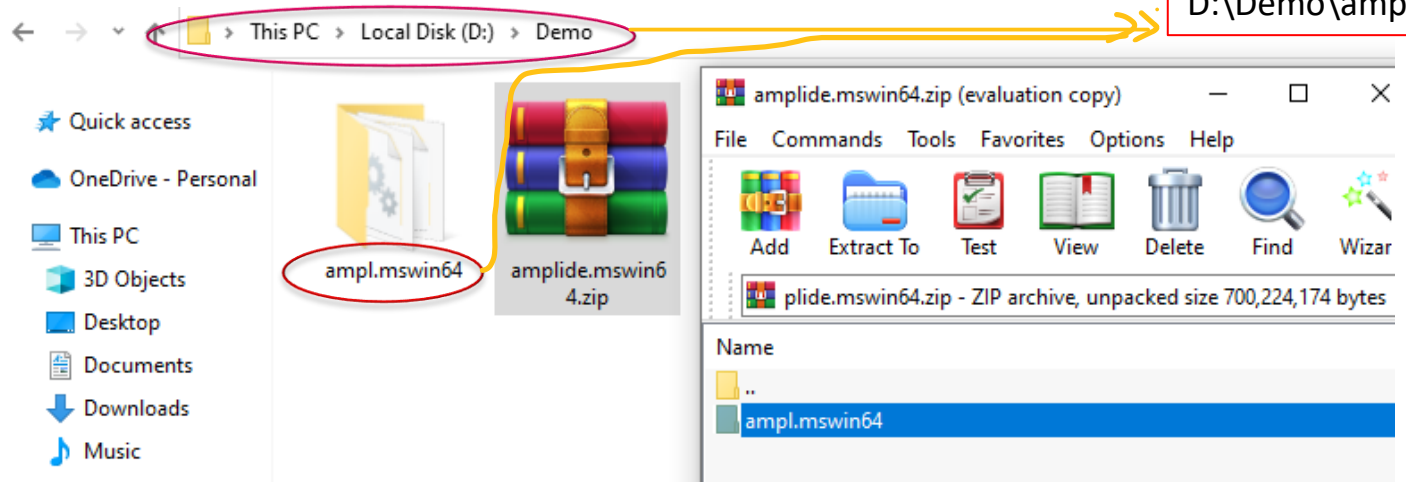
If OS is 64 or 32 bit – Windows, download links are here, respectively:

https://ampl.com/demo/ampl.mswin64.zip

https://ampl.com/demo/ampl.mswin32.zip

- Unzip (extract the folder: ampl.mswin64) from the zip file.

Note that amplPathDir is:
D:\Demo\ampl.mswin64



1.

Installation - (Install the API)

2. - From the webpage: https://ampl.com/products/api/

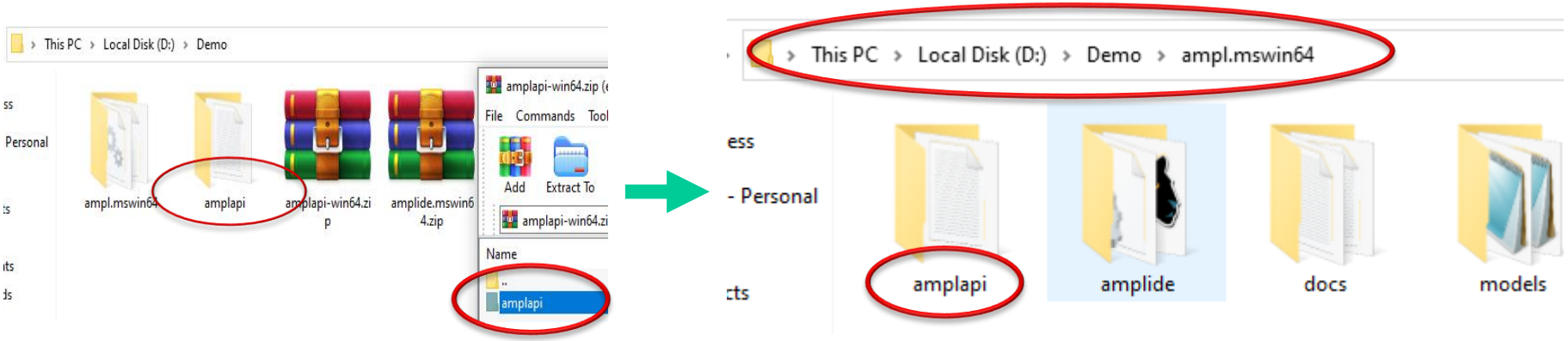
choose the download link for the API based on the OS and programming language.

Here are links of API for (C++, C#, Java, &) MATLAB APIs in Windows 64/32 bit respectively:

https://ampl.com/dl/API/latest/amplapi-win64.zip

https://ampl.com/dl/API/latest/amplapi-win32.zip

- Unzip (extract the folder: amplapli) from the zip file and then move to ampl.mswin64 folder.

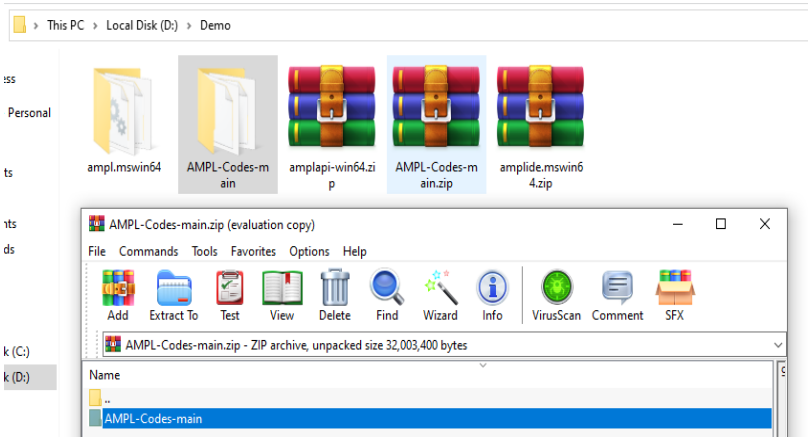


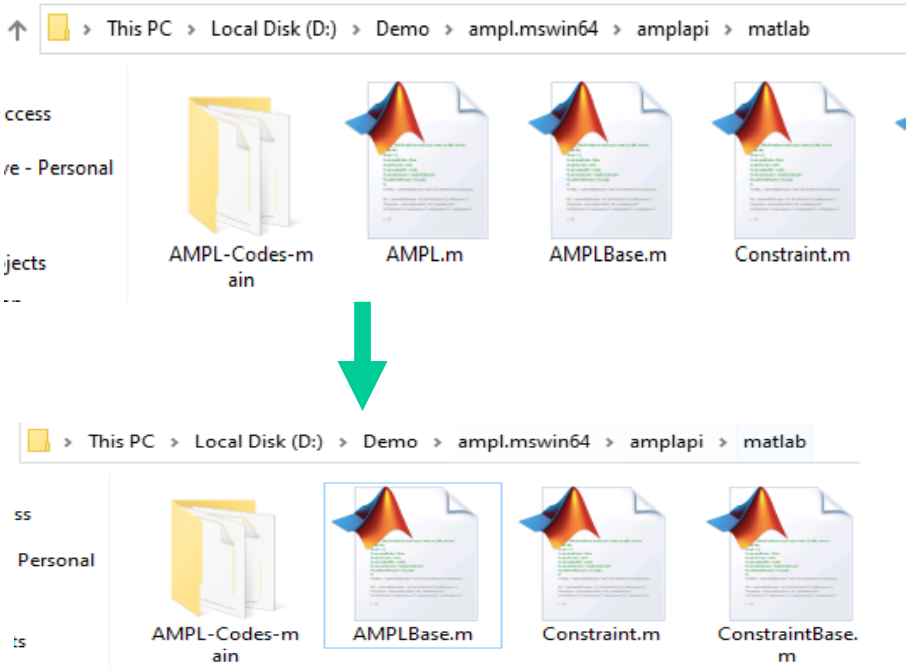
Installation - 3

3. - install the repository: AMPL-codes from the link:

https://github.com/Anas-Abdelkarim/AMPL-Codes

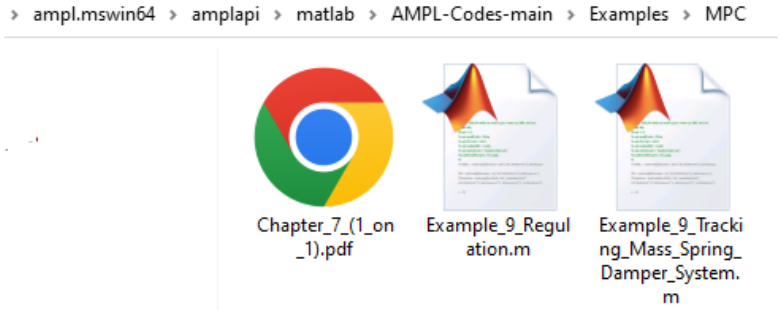
- unzip (extract the folder: AMPL-Codes-main) and move to: *ampl.mswin64\amplapi*
- Navigate to *ampl.mswin64\amplapi\matlab* and delete the file AMPL.m.





Installation – Run the Example

- To run the examples
- Navigate to *ampl.mswin64\amplapi\matlab\AMPL-Codes-main\Examples*
- For examples on model predictive control MPC run Example_9_Regulation.m and Example_9_Tracking_Mass_Spring_Damper_System.m inside of MPC folder



To run example for optimization problem with binary variables run Example_optimal_scheduling.m inside of Optimal-scheduling_example folder

Note: the students can request 30-day trail for unlimited decision variables using the link: https://ampl.com/products/ampl/ampl-for-students/#Trial

Run AMPL (Without installation) – NEOS server

Run AMPL

- **AMPL on NEOS server:**
  - It is free internet-based service for solving OP
  - It is hosted by the Unvisited of Wisconsin, and provides access to more than 60 well-established solvers
  - The submitted tasks (jobs) to the NEOS sever run on distribute high-performance computers.

1. Go to: https://neos-server.org/neos/solvers/index.html

2. Choose the target solver and click on AMPL

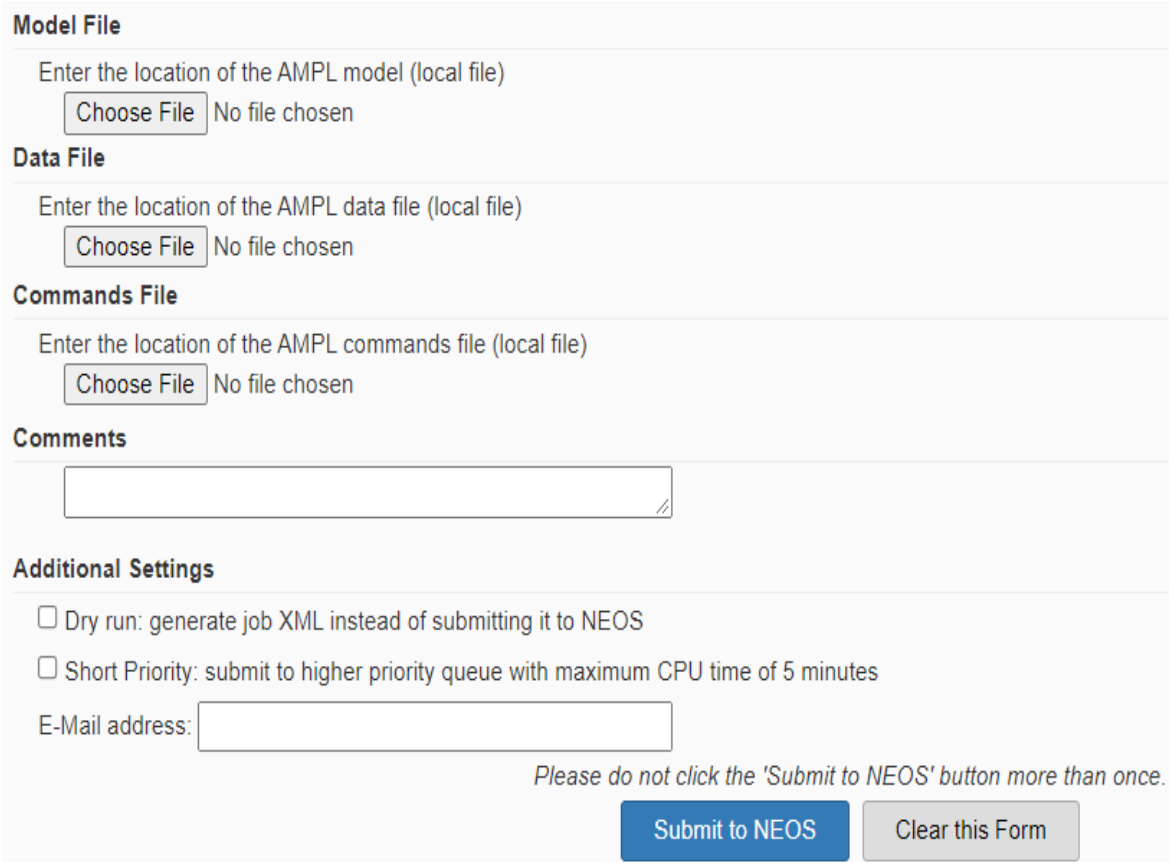
Mixed Integer Linear Programming

- Cbc [AMPL] [GAMS] [MPS]
- COPT [AMPL] [GAMS] [LP] [MPS] [NL]
- CPLEX [AMPL] [GAMS] [LP] [MPS] [NL]
- feaspump [AMPL] [CPLEX] [MPS]
- FICO-Xpress [AMPL] [GAMS] [MOSEL] [MPS] [NL]
- Gurobi [AMPL] [GAMS] [LP] [MPS] [NL]
- MINTO [AMPL]
- MOSEK [AMPL] [GAMS] [LP] [MPS] [NL]

Run AMPL (Without installation) – NEOS server

Run AMPL

- **AMPL on NEOS server:**
  3. Upload AMPL files, fill the email
  4. Click on Submit to NEOS

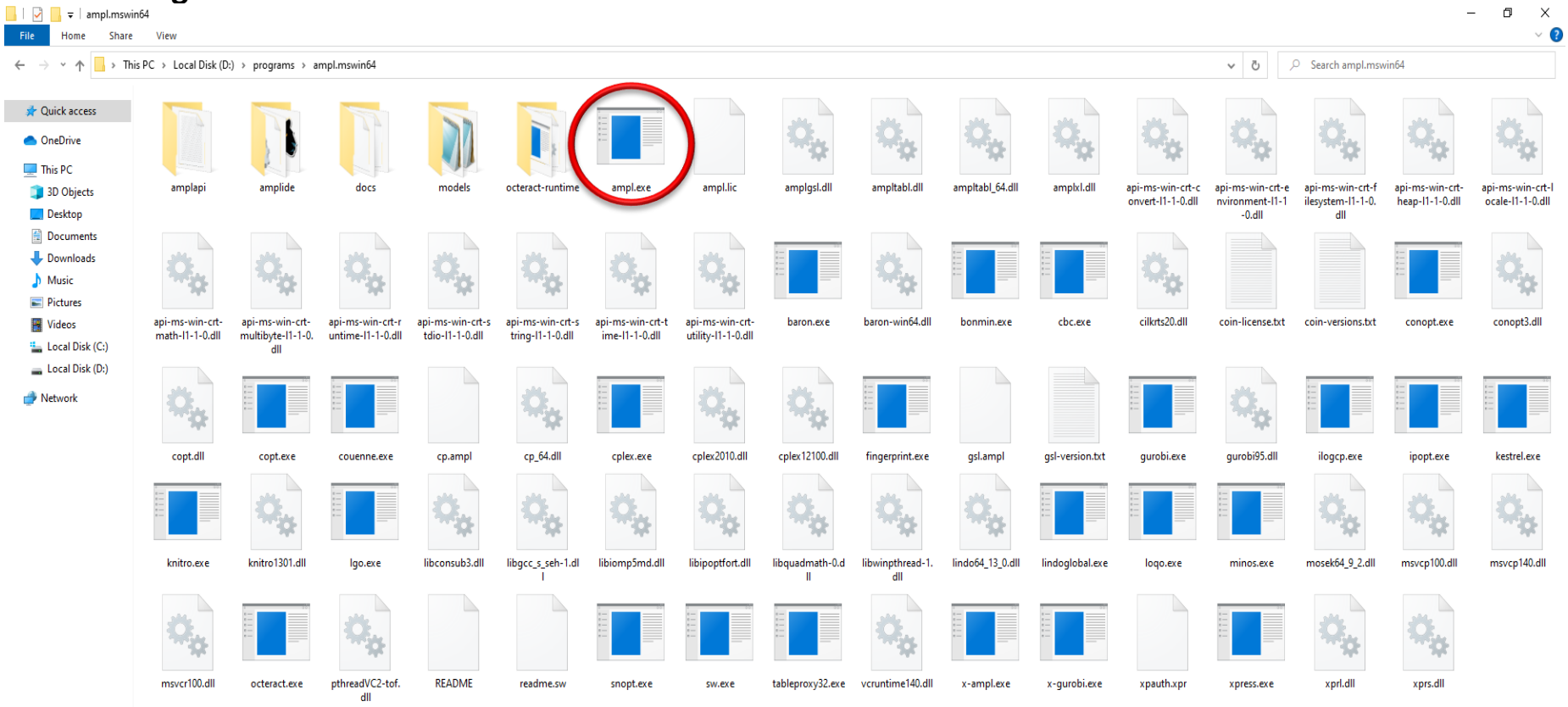


- **Remark:**
  - Run Should not have load *.mod, load *.dat, and option solver * code lines.

Run AMPL - form Terminal

Run AMPL

- **Using AMPL Terminal:**



Run AMPL - form Terminal

Run AMPL

- **Using AMPL Terminal:**

D:\programs\ampl.mswin64\ampl.exe

```
ampl: reset;
ampl: ##########################
ampl: model;
ampl: param a;
ampl: var x; # the decision variable
ampl: minimize cost: x**2;
ampl: s.t.              restriction: x >= a;
ampl: ##########################
ampl: data;
ampl data: param a := 2;
ampl data: ##########################
ampl data: solve;
MINOS 5.51: optimal solution found.
0 iterations, objective 4
Nonlin evals: obj = 3, grad = 2.
ampl: expand cost, restriction;
minimize cost:
        x^2;

subject to restriction:
        x >= 2;

ampl: display x, a;
x = 2
a = 2
```

$$a = 2$$

$$\underset{x}{\text{minimiz}}\ x^2$$

$$\text{subject to } x \geq a$$

```
reset;
##############################
model;
param a;
var x; # the decision variable
minimize cost: x**2;
s.t.          restriction: x >= a;
##############################
data;
param a := 2;
##############################
solve;
expand cost, restriction;
display x, a;
##############################
option solver cplex;
solve;
display x;
##############################
option solver ipopt;
solve;
display x;
```

Run AMPL - form Terminal

```
D:\programs\ampl.mswin64\ampl.exe
******************************************************************************
This program contains Ipopt, a library for large-scale nonlinear optimization.
 Ipopt is released as open source code under the Eclipse Public License (EPL).
         For more information visit http://projects.coin-or.org/Ipopt
******************************************************************************

This is Ipopt version 3.12.13, running with linear solver mumps.
NOTE: Other linear solvers might be more efficient (see Ipopt documentation).

Number of nonzeros in equality constraint Jacobian...:        0
Number of nonzeros in inequality constraint Jacobian.:        0
Number of nonzeros in Lagrangian Hessian.............:        1

Total number of variables............................:        1
                     variables with only lower bounds:        1
                variables with lower and upper bounds:        0
                     variables with only upper bounds:        0
Total number of equality constraints.................:        0
Total number of inequality constraints...............:        0
        inequality constraints with only lower bounds:        0
   inequality constraints with lower and upper bounds:        0
        inequality constraints with only upper bounds:        0

iter    objective    inf_pr   inf_du lg(mu)  ||d||  lg(rg) alpha_du alpha_pr  ls
   0  4.0803999e+00 0.00e+00 3.04e+00  -1.0 0.00e+00    -  0.00e+00 0.00e+00   0
   1  4.1177774e+00 0.00e+00 1.85e-02  -1.0 1.85e-02    -  1.00e+00 5.00e-01f  2
   2  4.0049122e+00 0.00e+00 2.83e-08  -2.5 2.80e-02    -  1.00e+00 1.00e+00f  1
   3  4.0001532e+00 0.00e+00 1.50e-09  -3.8 1.19e-03    -  1.00e+00 1.00e+00f  1
   4  4.0000018e+00 0.00e+00 1.84e-11  -5.7 3.79e-05    -  1.00e+00 1.00e+00f  1
   5  3.9999999e+00 0.00e+00 2.49e-14  -8.6 4.61e-07    -  1.00e+00 1.00e+00f  1

Number of Iterations....: 5

                                   (scaled)                 (unscaled)
Objective...............:   3.9999999225063294e+00    3.9999999225063294e+00
Dual infeasibility......:   2.4868995751603507e-14    2.4868995751603507e-14
Constraint violation....:   0.0000000000000000e+00    0.0000000000000000e+00
Complementarity.........:   2.5063293419916432e-09    2.5063293419916432e-09
Overall NLP error.......:   2.5063293419916432e-09    2.5063293419916432e-09


Number of objective function evaluations             = 11
Number of objective gradient evaluations             = 6
Number of equality constraint evaluations            = 0
Number of inequality constraint evaluations          = 0
Number of equality constraint Jacobian evaluations   = 0
Number of inequality constraint Jacobian evaluations = 0
Number of Lagrangian Hessian evaluations             = 5
Total CPU secs in IPOPT (w/o function evaluations)   =      0.011
Total CPU secs in NLP function evaluations           =      0.000

EXIT: Optimal Solution Found.

Ipopt 3.12.13: Optimal Solution Found
```

$a = 2$

$$\underset{x}{minimiz}\ x^2$$

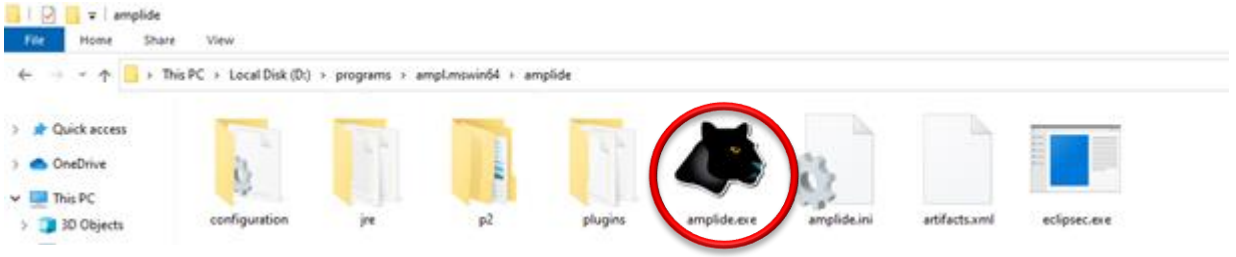
$$subject\ to\ x \geq a$$

```
reset;
############################
model;
param a;
var x; # the decision variable
minimize cost: x**2;
s.t.          restriction: x >= a;
############################
data;
param a := 2;
############################
solve;
expand cost, restriction;
display x, a;
############################
option solver cplex;
solve;
display x;
############################
option solver ipopt;
solve;
display x;
```

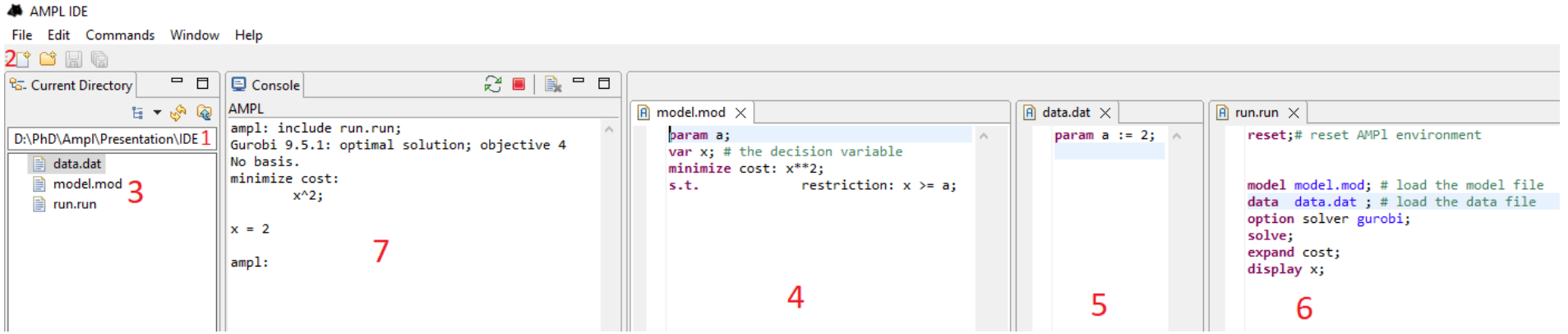
Run AMPL – using IDE

Run AMPL

- **Using AMPL IDE:**



$$a = 2$$
$$\underset{x}{\text{minimiz}}\ x^2$$
$$subject\ to\ x \geq a$$



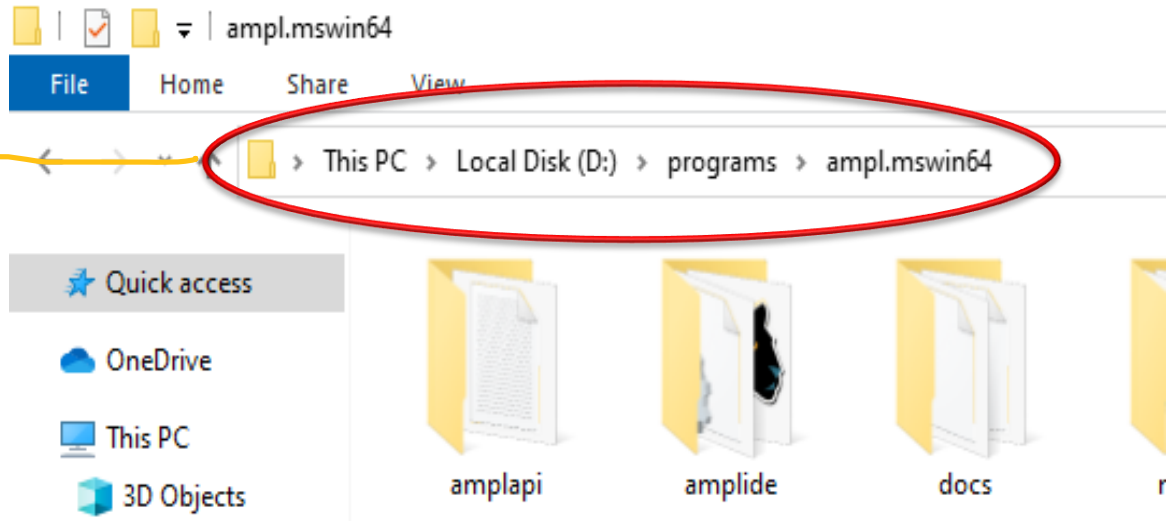
Run AMPL – in Matlab

Run AMPL

- **AMPL in Matlab (after installation of API)**

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%% AMPL Environment %%%%%%%%%%%%%%%%%%%%%%%%%%%%
amplPathDir = 'D:\programs\ampl.mswin64';
%% add path for the customized codes
addpath(amplPathDir+ "\amplapi\matlab\customizedCodes")
run(amplPathDir + "\amplapi\matlab"+ "\setUp.m")  % setup API for matlab
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

$$a = 2$$
$$\underset{x}{\text{minimiz}}\ x^2$$
$$subject\ to\ x \geq a$$



Run AMPL – in Matlab

Run AMPL

- **AMPL in MATLAB**

```
% run AMPL
ampl = AMPL(amplPathDir);
% model
ampl.eval('model; param a; var x;')
ampl.eval('minimize cost: x**2;')
ampl.eval('restriction: x >= a;')
%data
ampl.eval('data; param a:=2;')
% Choose the solver
ampl.eval('option solver gurobi;')
% solve the optimization problem
ampl.solve()
% extract the results:
[value, astatus,result,exitcode,message,expand] = ampl.getObj('cost');
x = ampl.getVarValue('x');
a = ampl.getParamValue('a');
% close AMPL
ampl.close
```

$$a = 2$$
$$\underset{x}{\text{minimiz}}\; x^2$$
$$subject\ to\ x \geq a$$

Examples – Mixed Integer Programming

- **Optimal Control: Mixed Integer Programming**
  - Abdelkarim, A. and Zhang, P., 2020, September. Optimal Scheduling of Preventive Maintenance for Safety Instrumented Systems Based on Mixed-Integer Programming. In *International Symposium on Model-Based Safety and Assessment* (pp. 83-96). Springer, Cham.

For more details, go to folder:

amplPathDir\amplapi\matlab\AMPL-Codes-main\Examples\Optimal-scheduling_example

Examples – Mixed Integer Programming

- **Optimal Control: Mixed Integer Programming**
  - Abdelkarim, A. and Zhang, P., 2020, September. Optimal Scheduling of Preventive Maintenance for Safety Instrumented Systems Based on Mixed-Integer Programming. In *International Symposium on Model-Based Safety and Assessment* (pp. 83-96). Springer, Cham.

For more details, go to folder:

amplPathDir\amplapi\matlab\AMPL-Codes-main\Examples\Optimal-scheduling_example

Example – (MODEL)

- **Optimal Control: Mixed Integer Programming**

$$\min_{\alpha(k),\gamma(k),p(k)} \quad \frac{1}{N}\sum_{k=0}^{N-1}(p_3(k)+p_4(k)+p_5(k)+p_6(k))$$

$$\text{s.t.} \quad p(k+1) = \gamma(k)(Qp(k)) + \alpha(k)(Q(M_A p(k)),$$

$$\sum_{k=0}^{N-1}\alpha(k) \le n_f, \qquad \alpha(k) \in \{0,1\}$$

$$\alpha(k)+\gamma(k) = 1, \qquad k = 0,1,\cdots,N-1$$

$$p(0) = p_0,$$

```
param N , integer                          ; # horizon length
var alpha { k in 0..N-1 } , binary         ; # full maintenance mode
var gamma { k in 0..N-1 } , binary         ; # dynamics mode
var p {i in 1..6 , k in  0..N  } >= 0      ; # states vector
param n_f                                  ; # num. of full maintenance
param initial_values {j in 1 ..6}          ; # initial values
param Q {i in 1..6, j in 1..6}             ; # system matrix
param Ma {i in 1..6, j in 1..6}            ; # full maintenance matrix

 minimize cost_function:
1/(N)*sum { k in 0..N-1} ( p[3,k]  +   p[4,k]  +  p[5,k]  +   p[6,k]  );

dynamics { i in 1..6,k in 0..N-1 } :
  p[i,k+1] =     (gamma[k] * sum { c in 1..6}  Q[i,c]  * p[c,k]) +
(alpha[k] * sum { c in 1..6} Q[i,c]*sum { j in 1..6}  Ma[c,j]*p[j,k]);

full_maintenance_active:          sum { j in 0..N-1} alpha [j] <= n_f;

one_active { k in 0..N-1}  :                    gamma[k] + alpha[k]  = 1;

subject to  initial_values_constraints {i in 1 .. 6}:
                                  p[i,0] = initial_values[i];
```

Examples – Example 1 (MODEL)

- **In MATLAB**

```
ampl.defineParam("N",1,"integer")
ampl.defineVar("alpha gamma ","0..N","binary")
ampl.defineVar("p",[6,"0..N"],">=0")
ampl.defineParam("n_f")
ampl.defineParam("Q Ma",[6,6])
ampl.defineParam("initial_values ",6)

cost_function = ['  1/(N)*sum { k in 0..N-1} ( ' ...
                 ' p[3,k] + p[4,k+1] + p[5,k] + x[6,k+1] ' ];
ampl.defineObj("min","cost_function",cost_function);

dynamics = ['  p[i,k+1] = ' ...
'  (gamma[k] * sum { c in 1..6} Q[i,c] * p[c,k]) + ' ...
'  (alpha[k] * sum { c in 1..6} Q[i,c]*sum { j in 1..6} Ma[c,j]*p[j,k]) ' ];
ampl.defineCons("dynamics",dynamics,"i in 1..6, k in 0..N-1" );

full_maintenance_active= 'sum { j in 0..N-1} alpha[j] <= n_f';
ampl.defineCons("full_maintenance_active",full_maintenance_active)

one_active =  ' gamma[j]  + alpha[j] = 1' ;
ampl.defineCons("one_active",one_active,"j in 0..N-1")
ampl.defineCons("initial_values_constraints", 'x[i,0] = initial_values[i]' ,"i in 1..6")
% Note: ampl.defineParam, defineVar, defineCons and
% ampl.defineCons are customized codes
```

```
param N , integer                       ; # horizon length
var alpha { k in 0..N-1 } , binary      ; # full maintenance mode
var gamma { k in 0..N-1 } , binary      ; # dynamics mode
var p {i in 1..6 , k in  0..N  } >= 0   ; # states vector
param n_f                               ; # num. of full maintenance
param initial_values {j in 1 ..6}       ; # initial values
param Q {i in 1..6, j in 1..6}          ; # system matrix
param Ma {i in 1..6, j in 1..6}         ; # full maintenance matrix

 minimize cost_function:
1/(N)*sum { k in 0..N-1} ( p[3,k]  +   p[4,k]  +  p[5,k]  +   p[6,k]  );

dynamics { i in 1..6,k in 0..N-1 } :
  p[i,k+1] =     (gamma[k] * sum { c in 1..6}  Q[i,c]  * p[c,k]) +
(alpha[k] * sum { c in 1..6} Q[i,c]*sum { j in 1..6}  Ma[c,j]*p[j,k]);

full_maintenance_active:          sum { j in 0..N-1} alpha [j] <= n_f;

one_active { k in 0..N-1}  :                  gamma[k] + alpha[k]  = 1;

subject to  initial_values_constraints {i in 1 .. 6}:
                                  p[i,0] = initial_values[i];
```

Examples – Example 1 (DATA)

- **Optimal control: mixed integer programming**

$$\min_{\alpha(k),\gamma(k),p(k)} \quad \frac{1}{N}\sum_{k=0}^{N-1}(p_3(k)+p_4(k)+p_5(k)+p_6(k))$$

$$\text{s.t.} \quad p(k+1)=\gamma(k)(Qp(k))+\alpha(k)(Q(M_A p(k)),$$

$$\sum_{k=0}^{N-1}\alpha(k)\le n_f, \qquad \alpha(k)\in\{0,1\}$$

$$\alpha(k)+\gamma(k)=1, \qquad k=0,1,\cdots,N-1$$

$$p(0)=p_0,$$

```
param n_f   := 1 ;
param initial_values      :=  1  1
                              2  0
                              3  0
                              4  0
                              5  0
                              6  0 ;
param N := 131400;          # 15 years
param Q :  1      2       3       4       5       6 :=
1          1    0.005   0.001     0       0       0
2          0      1       0       0       0       0
3          0      0       1       0       0       0
4          0      0       0       1       0       0
5          0      0       0       0       1       0
6    0     0      0       0       0       1;

param Ma :  1 2 3 4 5 6  :=
        1   1 0 0 0 0 0
        2   0 1 0 0 0 0
        3   0 0 1 1 1 0
        4   0 0 0 0 0 0
        5   0 0 0 0 0 0
        6   0 0 0 0 0 1;
```



Examples – Example 1 (DATA)

- **In MATLAB**

```
n_f= 1 ;
initial_values = [ 1 0 0 0 0 0];
N = 131400;

Q =  [1    0.005   0.001     0       0       0
      0     1       0        0       0       0
      0     1       0        0       0       0
      0     0       0        1       0       0
      0     0       0        0       1       0
      0     0       0        0       0       1];

Ma = [ 1 0 0 0 0 0;0 1 0 0 0 0;0 0 1 1 1 0;0 0 0 0 0 0;
       0 0 0 0 0 0 ; 0 0 0 0 0 1];

ampl.assignParam("N n_f", [N n_f])
ampl.assignParam("initial_values",initial_values)
ampl.assignParam("Ma",Ma)
ampl.assignParam("Q",Q)

% Note: ampl.defineParam, defineVar, defineCons and
% ampl.defineCons are customized codes
```

```
param n_f   := 1 ;
param initial_values      := 1 1
                             2 0
                             3 0
                             4 0
                             5 0
                             6 0 ;
param N := 131400;          # 15 years
param Q :   1     2       3       4       5       6 :=
1           1   0.005   0.001     0       0       0
2           0     1       0       0       0       0
3           0     0       1       0       0       0
4           0     0       0       1       0       0
5           0     0       0       0       1       0
6   0       0       0       0       0       1;

param Ma :  1 2 3 4 5 6  :=
            1  1 0 0 0 0 0
            2  0 1 0 0 0 0
            3  0 0 1 1 1 0
            4  0 0 0 0 0 0
            5  0 0 0 0 0 0
            6  0 0 0 0 0 1;
```

Examples – Example 1 (SOLVE)

- **In MATLAB**

```
%% solve the optimization problem
ampl.solve


%% Harvest the results
% check the value of N
ampl.getParamValue("N")
ampl.getParamValue("initial_values")
ampl.getParamValue("Ma")
[gamma, index,rowdata] = ampl.getVarValue("gamma");
alpha = ampl.getVarValue("alpha");
p = ampl.getVarValue("p");




% objective function
[value, astatus,result,exitcode,message,expandCost] = ampl.getObj('cost_function')
```

Helpful functions

Customized codes for AMPL in MATLAB:

- expand(input): results in a string array of a matrix, vector, or scalar elements
- Examples:

Line command	expand('Q[1..3,1..2]')	expand('y[0..2,k]')	expand('s[1..1]')	expand('v.T[0..%u]',1)
Data Type Output	3×2 string array	3×1 string array	String	1×2 string array
Output	"Q[1,1]" "Q[1,2]" "Q[2,1]" "Q[2,2]" "Q[3,1]" "Q[3,2]"	"y[0,k]" "y[1,k]" "y[2,k]"	"s"	"v[0]" "v[1]"



Helpful functions

Customized codes for AMPL in MATLAB:

- multiply(A,B): results in a string array of a product of A and B
- Examples:

Line command	Data Type Output	Output
multiply('x.T[1..2,k] ','Q[1..2,1..2] ')	1×2 string array	"(x[1,k]*Q[1,1]+ x[2,k]*Q[2,1])" "(x[1,k]*Q[1,2]+ x[2,k]*Q[2,2])"
multiply('x.T[1..2,k] ','Q[1..2,1..2] ', 'x[1..2,k] ',)	string	"((x[1,k]*Q[1,1]+ x[2,k]*Q[2,1])*x[1,k]+ (x[1,k]*Q[1,2]+ x[2,k]*Q[2,2])*x[2,k])"



References

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2. https://www.youtube.com/watch?v=hrqsflMu4z8&ab_channel=AMPLOptimization
3. https://neos-server.org/neos/
4. Abdelkarim, A. and Zhang, P., 2020, September. Optimal Scheduling of Preventive Maintenance for Safety Instrumented Systems Based on Mixed-Integer Programming. In *International Symposium on Model-Based Safety and Assessment* (pp. 83-96). Springer, Cham.